12TH BIOLOGY · CHAPTER 19 · CHAPTER TEST

Chromosome and DNA

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Biology	**Total Marks:** 26
Chapter: 19	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Which of the following looks like beads on a string?

(a) Chromosomes (b) Chromatin
(c) Heterochromatin (d) Nucleosomes

2. Which enzyme joins the bits of DNA?

(a) DNA polymerase (b) DNA ligase
(c) Endonuclease (d) Primase

3. Which subunit of RNA polymerase is required to initiate transcription?

(a) Sigma (σ) (b) Alpha (α)
(c) Omega (ω) (d) Beta (β)

4. The synthesis of protein chains is initiated with the amino acid:

(a) Arginine (b) Serine
(c) Methionine (d) Valine

5. The condition sickle cell anaemia is due to:

(a) Deletion type point mutation (b) Substitution type point mutation
(c) Insertion type point mutation (d) Chromosomal rearrangement

6. The number of chromosomes in Turner syndrome is:

(a) 44 (b) 46
(c) 45 (d) 47

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. What are the four types of chromosomes on the basis of position of centromere?
ii. What is transformation?
iii. State the functions of DNA polymerase I, II and III.
iv. What do you mean by “degeneracy of genetic code”?
v. Differentiate between intron and exon.
vi. What is non-disjunction?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Describe the events of the process of DNA replication.
(b) Describe the symptoms, causes and treatments of Down syndrome.

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